

Phase 8: Data Management & Deployment

Project Title: Student Success & Career Guidance Portal

8.1 Data import — Data Import Wizard (simple objects)

Prepare: Students.csv example headers:

FirstName,LastName,Email,Student_ID__c,AccountName,GPA__c

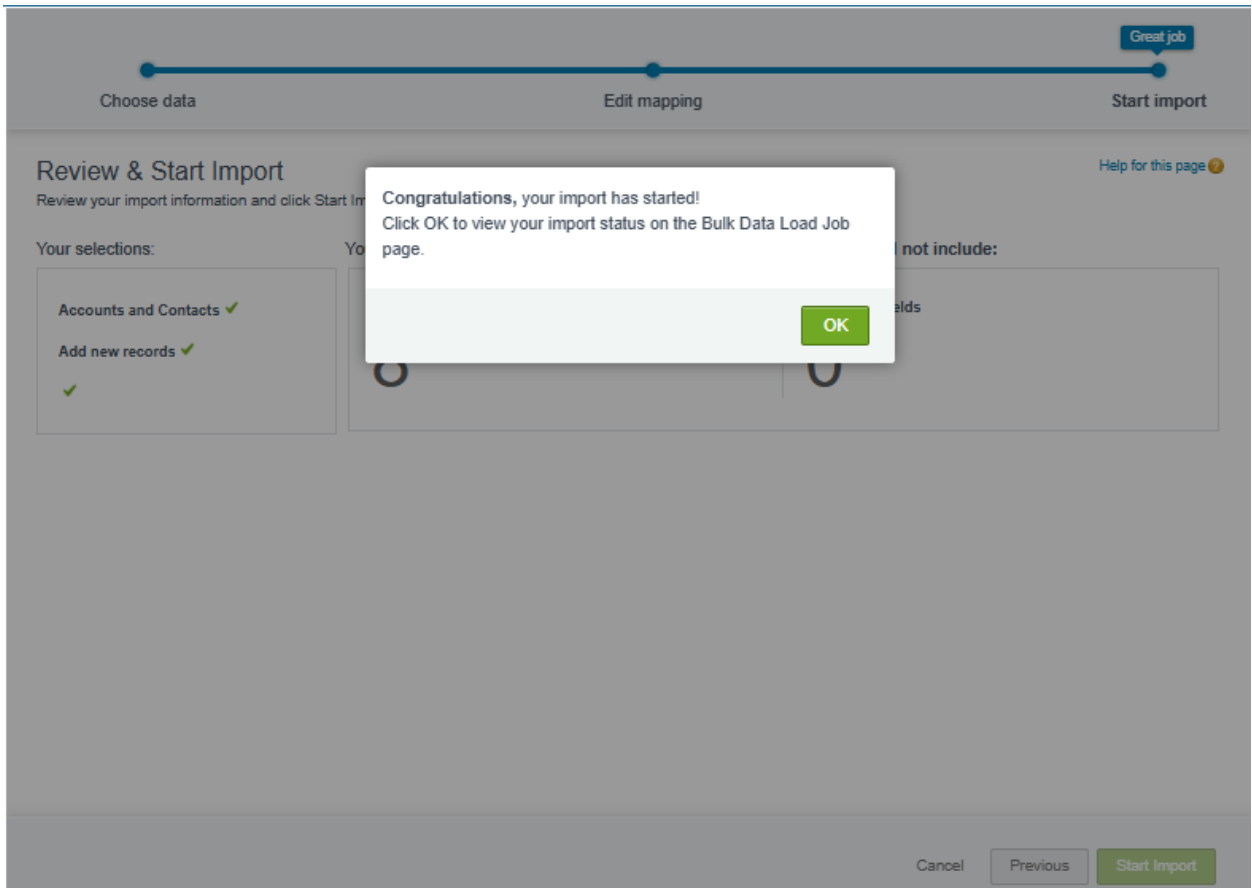
Swayam,Tamrakar,swayam@example.com,STU001,XYZ Engineering College,8.2

Note: You can import Contacts and specify Account by **Account Name** (Data Import Wizard will try to match or create).

Steps

1. In Salesforce: click  → **Setup**.
2. Quick Find → **Data Import Wizard** → **Launch Wizard**.
3. Choose the object group (e.g., **Standard Objects** → Accounts and Contacts).
4. Select operation: **Add new records** (or Update existing if you include Salesforce IDs).
5. Choose CSV file (Students.csv) and follow the mapping screen: map CSV columns to Salesforce fields.
 - Map AccountName → Account Name (if importing Contacts tied to Account).

6. Review and click **Start Import**.



7. After run completes, check the email notification and import results page for success/error counts.

8. Fix errors (download error CSV), correct CSV, re-import failed rows.

Bulk Data Load Job

750gL00000E24CO

View the details of a bulk data load job.

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Bulk Data Load Job Detail

Reload

Job ID	750gL00000E24CO	Job Type	Bulk V1	Status	Closed
Submitted By	Swayam Tamrakar	Operation	Insert	Total Processing Time (ms)	806
Start Time	9/24/2025, 2:21 AM PST	Queued Batches	0	API Active Processing Time (ms)	601
End Time	9/24/2025, 2:21 AM PST	In Progress Batches	0	Apex Processing Time (ms)	170
Time to Complete (hh:mm:ss)	00:01	Completed Batches	1		
Object	Account	Failed Batches	0		
External ID Field		Progress	100%		
Content Type	CSV	Records Processed	100		
Concurrency Mode	Parallel	Records Failed	0		
API Version	64.0	Retries	0		

Reload

Batches

act	View Result	Batch ID	Start Time	End Time	Total Processing Time (ms)	API Active Processing Time (ms)	Apex Processing Time (ms)	Records Processed	Records Failed	Retry Count	State Message	Status
	View Result	751gL00000BbYSe	9/24/2025, 2:21 AM	9/24/2025, 2:21 AM	806	601	170	100	0	0		Completed

◀

▶

Salesforce import of "Students_100 (1).csv" has finished. 100 rows were processed. External Inbox x



noreply@salesforce.c... 14:52 (0 minutes ago) ☆ ↩ ⋮
to me ▾

Your Accounts and Contacts imports are complete. Here are your results:

Accounts Created: 100

Accounts Updated: 0

Accounts Ignored: 0 (We ignored updates that we couldn't match to an existing record.)

Accounts Failed: 0 (We couldn't import these due to errors.)

Processed job information for imported Accounts: <https://orgfarm-081ad31f5a-dev-ed.develop.my.salesforce.com/750gL00000E24COQAZ?fromEmail=1>

Tips

- For initial data load, import Accounts first (Colleges/Companies), then Contacts (Students) referencing accounts.
- If you plan to use Data Loader later, you can import basic data with the Wizard and use Data Loader for complex objects.

8.1 Data import — Data Loader (bulk, lookups, upsert)

Best for: bulk loads, custom objects, junction objects, upsert using External IDs.

Install & Login

1. Install Data Loader and open it.
2. Click **Login** → choose **Username-Password** (enter security token if required)
OR use **OAuth** flow (recommended).
 - For Developer Org, if using username&password, append your security token to password.

Common operations: Insert, Update, Upsert, Export.

Example: Load Student_Skill__c (junction)

Problem: Student_Skill__c has lookup to Contact and Skill. You likely don't have Salesforce Ids initially.

Use IDs

1. Export Contacts (after Contacts import) to get Contact IDs: Setup → Data Loader → Export → select Contact → export Id, Email, Student_ID__c to CSV.

2. Prepare Student_Skill.csv:

Student__c,Skill__c,Proficiency__c

0033t00002ABCDE, a0H3t00000Skill1,Intermediate

0033t00002FGHIj, a0H3t00000Skill2,Beginner

3. In Data Loader: **Insert** → select Student_Skill__c → choose CSV → Map fields (or auto map) → Finish → check success/error files.

Option B — Use External ID upsert (no Salesforce IDs)

1. Create an External ID field on Contact:
 - Setup → Object Manager → Contact → Fields & Relationships → **New** → Text → Student_External_Id__c and check **External ID**.

2. Ensure Skill__c has External ID (or create Skill_Code__c External ID).

3. Prepare Student_Skill_Upsert.csv:

Student_External_Id__c,Skill_Code__c,Proficiency__c

STU001,SKL_JAVA,Intermediate

STU002,SKL_PYTHON,Advanced

4. In Data Loader: choose **Upsert** → select Student_Skill__c → for the lookup fields use the ExternalId mapping: map Student_External_Id__c → Student__r.Student_External_Id__c and Skill_Code__c → Skill__r.Skill_Code__c (Data Loader supports parent external id mapping using relationship notation).
5. Run import and review success/error files.

Tips

- Always **export a small sample** first to confirm mappings.
 - Keep the **success.csv** and **error.csv** produced by Data Loader — they tell you row-level outcome and error messages.
 - For large loads split into batches < 10k if necessary. Use Bulk API mode in Data Loader for huge data.
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Recommended load order (best practice)

1. Accounts (Colleges / Companies)
 2. Contacts (Students) — include AccountId or Account Name
 3. Skills (Skill__c)
 4. Junctions Student_Skill__c (use Contact Id or External ID upsert)
 5. Internships (Internship__c) — include Student__c lookup and Company Account lookup
 6. Placements (Placement__c)
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8.2 Duplicate Rules — create matching & duplicate rules (step-by-step)

Goal: Prevent or alert on duplicate Student (Contact) records by Email or Student_ID__c.

Create Matching Rule

1. Setup → Quick Find → **Matching Rules** → **New Rule**.
 - Select **Contact** as Object.
 - Rule Name: Contact_Match_By_Email_Or_StudentID.
 - Add Matching Criteria:
 - Field: Email Match Method: Exact
 - OR add Student_ID__c Match Method: Exact (if you have Student_ID__c).

2. Save and **Activate** the matching rule (must be active to use).

Matching Rules

Matching Rule
Contact_Match_By_Email_Or_StudentID
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Matching Rule Detail [Delete](#) [Clone](#) [Deactivate](#)

Object	Contact
Rule Name	Contact_Match_By_Email_Or_StudentID
Unique Name	Contact_Match_By_Email_Or_StudentID
Description	
Matching Criteria	Contact: Email EXACT MatchBlank = FALSE
Status	Active
Created By	Swayam Tamrakar, 9/23/2025, 6:03 AM
Modified By	Swayam Tamrakar, 9/23/2025, 6:03 AM

Create Duplicate Rule

1. Setup → Quick Find → **Duplicate Rules** → **New Rule** → choose **Contact**.
2. Rule Name: Contact_Duplicate_Rule_Students.
3. Under **Matching Rule**, select the matching rule you created.
4. Action on create/edit:
 - **Alert** (allow save but show alert to user) or **Block** (prevent save) — choose based on policy.
5. Under **Actions**, specify what to do on create/edit — you can create a report, allow users to bypass with reason, or block.

6. Save and **Activate** the duplicate rule.

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SETUP

Duplicate Rules

Contact Duplicate Rule

Contact_Duplicate_Rule_Students

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Duplicate Rule Detail

Edit

Delete

Clone

Deactivate

Rule Name	Contact_Duplicate_Rule_Students	Order	3 of 3 [Reorder]
Description			
Object	Contact		
Record-Level Security	Enforce sharing rules		
Action On Create	Allow	Operations On Create	☒ Alert ☒ Report
Action On Edit	Allow	Operations On Edit	☒ Alert ☒ Report
Alert Text	Use one of these records?		
Active	☒		
Matching Rule	☒ Contact Match By Email Or StudentID Mapped ☒	Matching Criteria	Contact: Email EXACT MatchBlank = FALSE
Conditions			
Created By	Swayam Tamrakar , 9/23/2025, 6:05 AM	Modified By	Swayam Tamrakar , 9/23/2025, 6:05 AM

Edit

Delete

Clone

Deactivate

Test

- Attempt to create a Contact with same email or Student_ID__c — verify the behavior (alert or block).

The screenshot shows a Salesforce 'New Contact' form. The contact is named Swayam Tamrakar, associated with XYZ Engineering College. The email field is highlighted in yellow and contains 'epic.orgfarm@salesforce.com'. A red error message box is overlaid on the form, stating: 'We hit a snag. You can't save this record because a duplicate record already exists. To save, use different information. View Duplicates'. At the bottom, there are buttons for 'Cancel', 'Save & New', and 'Save'. The GPA field is visible at the bottom left with the value '8.80'.

8.3 Backups — Data Export (manual & scheduled)

Manual Export (Developer Org)

1. Setup → Quick Find → **Data Export** → Click **Export Now**.
2. Choose objects to export (or select **Include all data**). For faster test, select Accounts, Contacts, Internship__c, Student_Skill__c, etc.

3. Click **Start Export**. Salesforce prepares a ZIP file and sends email when available. Download the ZIP and store it securely.

 **SETUP**
Data Export

Monthly Export Service [Help for this Page](#)

Data Export lets you prepare a copy of all your data in salesforce.com. From this page you can start the export process manually or schedule it to run automatically. When an export is ready for download you will receive an email containing a link that allows you to download the file(s). The export files are also available on this page for 48 hours, after which time they are deleted.

 **Next scheduled export:**
None

[Export Now](#) [Schedule Export](#)

Scheduled By	Swayam Tamrakar
Schedule Date	9/23/2025
Export File Encoding	ISO-8859-1 (General US & Western European, ISO-LATIN-1)

Action	File Name	File Size
download	WE_00Dgl000007QLcfUAG_1.ZIP	254.5K

Scheduled Export (if available in your edition)

1. On **Data Export** page → **Schedule Export** (if enabled): choose **Weekly** or **Monthly**, select day/time and objects. Save.

The screenshot shows the 'Schedule Data Export' configuration page in Salesforce. At the top, there's a 'SETUP' link and the 'Data Export' header. The main section is titled 'Schedule Data Export' and includes 'Save' and 'Cancel' buttons. Below this, there are several configuration options:

- Export File Encoding:** A dropdown menu set to 'ISO-8859-1 (General US & Western European, ISO-LATIN-1)'.
- Include images, documents, and attachments:** An unchecked checkbox.
- Include Salesforce Files and Salesforce CRM Content document versions:** An unchecked checkbox.
- Replace carriage returns with spaces:** A checked checkbox.
- Schedule Data Export:** A section containing:
 - Frequency:** Radio buttons for 'On day 1 of every month' (selected) and 'On the 1st Sunday of every month'.
 - Start:** A date field set to '9/24/2025' with a help icon.
 - End:** A date field set to '10/24/2025' with a help icon.
 - Preferred Start Time:** A dropdown menu set to '--None--'.
 - A note: 'Exact start time will depend on job queue activity.'

At the bottom, there's another 'Save' and 'Cancel' button. Below the main configuration area is the 'Exported Data' section, which includes a description and a list of data types to include in the export:

Select what type of information you would like to include in the export. The data types listed below use the Apex API names. If you are not familiar with these names, select Include all data for your export.

- ☒ Include all data
- ☐ Contract
- ☐ Asset
- ☐ Lead
- ☐ BusinessProcess
- ☐ Campaign
- ☐ CaseContactRole
- ☐ Order
- ☐ Account
- ☐ Partner
- ☐ NotificationMember
- ☐ CampaignMember
- ☐ CaseHistory2
- ☐ OrderItem
- ☐ Contact
- ☐ Product2
- ☐ UserRole
- ☐ Case
- ☐ CaseSolution

Tip: Always download and keep backup before any mass delete or destructive deploy.